Observatoire de Genève, UNIGE
Chemin Pegasi 51
1260 Versoix, Switzerland
Email: romain.meyer@unige.ch
https://rameyer.github.io/
ORCID ID: 0000-0001-5492-4522

Dr. Romain A. Meyer

RESEARCH INTERESTS

Using a wide range of observational probes and facilities from the optical and infrared to the millimeter domain, I aim to characterise the properties of galaxies in the first billion years of the Universe, their role in cosmic hydrogen reionisation, and their co-evolution with their central supermassive black holes. Starting in December 2026 I will lead an ERC-funded group at the University of Geneva aiming to transform the way we observe reionisation with JWST and next-generation VLT/ELT instruments.
Keywords: Cosmic Reionisation, First Galaxies & AGN/SMBH, Multi-wavelength

EMPLOYMENT

Assistant Professor December 2026 –
Department of Astronomy, University of Geneva, Switzerland

Maître-Assistant (Research & Teaching Fellow) April 2023 – November 2026
Department of Astronomy, University of Geneva, Switzerland
Group of Prof. Pascal Oesch

Postdoctoral Researcher October 2020 - March 2023
Max Planck Institute for Astronomy, Heidelberg, Germany
Group of Dr. Fabian Walter

EDUCATION

PhD in Astrophysics 2017 - 2020
University College London, United Kingdom
Thesis: *The Role of Galaxies and Quasars in Reionising the High Redshift Intergalactic Medium*
Supervisor: Prof. Richard S. Ellis
Thesis official electronic version: https://discovery.ucl.ac.uk/id/eprint/10112095/

MSc in Physics 2015 - 2017
Ecole Polytechnique Fédérale de Lausanne, Switzerland
Thesis: *PSF Interpolation via Artifical Neural Networks*
Supervisors: Prof. Frédéric Courbin, Dr. Thibault Kuntzer

BSc in Physics 2012 - 2015
Ecole Polytechnique Fédérale de Lausanne, Switzerland / Erasmus at Imperial College, London
Thesis: *Cross-calibration of the Herschel SPIRE instruments*
Supervisor: Prof. Dave Clements, Dr. Rosalind Hopwood

PUBLICATIONS

87 (14 first-author) refereed papers published in major astronomical journals (Nature, Nature Astronomy, A&A, MNRAS, ApJ, PASA, PASP, OJA) since 2018, totaling 4528 citations for an h-index of 36. A further 19 preprints under review, 2 proceedings publications and open source software releases. The full list of my publications can also be found on NASA ADS or on my ORCID profile.

GRANTS AND THIRD-PARTY FUNDING

ERC Starting Grant 2026 (2.27 M€) September 2026
ERC-STG-2026#101301766 "PostTenebrasLux" funding my group at UNIGE
NASA STScI Funding for JWST-GO Program #7208 (170'000 $) December 2025
Administered by US Admin Co-PI Dr. Christina Williams
SNSF Scientific Exchange (19'200 CHF) October 2024
Funding for the organisation of the Saas-Fee Advanced Course 2025
MERAC funding and travel award (2'400 CHF) May 2024
Funding for collaboration visits to support research projects on $z > 6$ quasar host galaxies

AWARDED TELESCOPE TIME

PI-led Programmes:

Career total of > 600 hours on major observational facilities (e.g. JWST, VLT, ALMA, Gemini, NOEMA) with $> 5 - 10$ oversubscription factors.

ALMA Band 1 - 29.7h 2026
"Weighing the Cold Molecular Gas Reservoirs of $z > 5$ Radio-Loud Quasars" (2026.1.00540.S)
JWST NIRSpec MSA - 108h 2026
"Mapping Reionisation in 3D in the COSMOS field (Cycle 5, #11371)"
VLT/FORS2 - 84h 2026
"Mapping Reionisation in 3D in the COSMOS field (117.29KC)"
ALMA Band 1 - 34.3h 2025
"Weighing the molecular gas reservoirs of UV-faint z>6 quasars" (2025.1.01221.S)
JWST NIRSpec IFU SURVEY - 236.2h 2025
"THRIFTY: The High-RedshIft FronTier surveY" (Cycle 4, #7208)
IRAM/NOEMA Band 3 - 30h 2024
"Characterising the ISM of a z=6.5 post-quasar galaxy" (W24EN)
ALMA Band 1 - 32h 2024
"A survey of cold molecular gas in $z \sim 6.5$ quasars and their companion galaxies" (2024.1.00106.S)
VLT/XShooter - 2.9h 2023
"Characterising a $z \sim 6$ gravitationally-lensed BAL quasar candidate" (113.26CY)
JWST NIRSpec IFU - 10.5h 2022
"Characterizing the Source of Ionising Photons in the Epoch of Reionization" (#3767)
Gemini (DDT) Flamingos-2 - 4.2h 2022
"Measuring the mass of the SMBHs powering two lensed $z \sim 6$ quasars" (DT-2022B-029)
NOEMA Band 3 - 12h 2022
"Disentangling the nature of JWST ultra-high-redshift candidates with NOEMA" (W22EG)
VLT/MUSE - 30h 2022
"A conclusive detection of galaxy overdensities around the first luminous quasars" (110.23UT)
ALMA Band 6 - 14.6h 2021
"The Large-Scale Environment of the First Quasars" (2021.1.01557.S)
ALMA Band 8/9 - 16.3h 2021
"Extreme Super-Eddington Star Formation in a Quasar Host at $z \sim 7$?" (2021.1.01350.S)
NTT/EFOSC - 4 nights 2019
"A search for the missing gravitationally-lensed $z \sim 6$ quasars" (0104.A-0662(A))

Selected Co-I Programmes:

ALMA LP - 113h 2025
"PHOENIX: the Emergence of Dust, Obscured Star Formation and ISM Physics at Cosmic Dawn" (PI: S. Shouws)

JWST LP #5893 - 266h 2024
"COSMOS-3D: A Legacy Spectroscopic/Imaging Survey of the Early Universe"
(PI: K. Kakiichi)
JWST #5645 - 389h 2024
"A 3D view of the first QSOs: A JWST/NIRSpec survey program" (PI: E. Farina)
VLT/MUSE LP - 147h 2023
"The cosmic Ecosystem of the first QSOs and Galaxies" (PI: E. Farina)
ALMA - 49h 2023
"Delivering the Needed Large Samples of Extremely High SFR Sources at $z > 6$
to Characterize Early Stellar+Black-Hole Growth" (PI: R. Bouwens)
JWST #2078 - 61h 2021
"A SPectroscopic survey of biased halos In the Reionization Era (ASPIRE)" (PI: F. Wang)

Additional Co-I Programmes: total of 409h with JWST, 332h with ALMA, 50.7h on the VLT (MUSE/ XShooter/FORS2), 22h on MEGARA/GTC, 15.6h on Subaru/PFS, 2×0.5 nights on Subaru/IRCS+LGS-AO, 6 nights on NTT, 92h on NOEMA, 38h on GEMINI GMOS, 19 orbits on HST, 333 ks on CHANDRA ACIS-S

STUDENTS SUPERVISION

Casey Hartmann, UNIGE MSc Semester Project Spring 2026

Alexandre Pozzi, UNIGE MSc Thesis Feb. 2025 - Jan 2026
Alexandre then moved to a Research Assistant position at George Mason University.

Lavinia Arpaia, UNIGE MSc Semester Project Spring 2024

Alexandre Pozzi, UNIGE MSc Semester Project Fall 2023

Xander Byrne, MPIA Summer Internship Summer 2022
1 publication accepted in MNRAS (link here). Xander then moved to a PhD position at the University of Cambridge.

THESIS JURIES AND EXAMINATIONS

External Examiner, MSc Thesis (A. Sawarkar), University of Geneva June 2026

External Examiner, Geneva Highschool (CECG De Staël) Final Exams June 2026

TEACHING

Guest Lecturer - Galaxies & Cosmology I & II (UNIGE) 2024 - present
MSc level Lectures on the Epoch of Reionisation / AGN / Observational Cosmology
Teaching Assistant (UCL) 2019
Classical Mechanics
Teaching Assistant (EPFL) 2016 - 2017
2nd year Physics Labs
Teaching Assistant (EPFL) 2013 - 2016
General Physics for Engineers

PROFESSIONAL SERVICE

Geneva Observatory - Postdoc Representative 2026

Referee for Science 2026 - present

COSMOS-3D Collaboration - IGM & Reionisation WG Lead 2025 - present

Expert Panel Member - ESO Observing Programmes Committee 2025-2026

Saas-Fee Advanced Course 2025 Organiser (website) 2025
EAS 2024 Special Session 1 Co-Organiser and Chair 2024
EPFL-UNIGE Journal Club Organiser 2023 - 2025
Galaxy Coffee Seminar Organiser, MPIA, Heidelberg 2021 - 2022
Walter Group Meetings Organiser, MPIA, Heidelberg 2021 - 2022
Referee for ApJ/AJ/ApJS 2021 - present
Referee for MNRAS 2020 - present
Astronomy PhD Student Representative, UCL 2017- 2020

MEMBERSHIP OF PROFESSIONAL SOCIETIES

Internationl Astronomical Union, Junior Membership 2026 - present
Swiss Society for Astronomy & Astrophysics 2023 - present
Royal Astronomical Society 2018 - 2020

OBSERVING AND TECHNICAL EXPERIENCE

- **Observing experience:** 3 nights on Keck/MOSFIRE, 3 nights on Keck/DEIMOS, 4 nights on NTT/ EFOSC2, 3 nights (remote) on MEGARA/GTC
- **Data reduction & analysis:** optical/NIR multi/long-slit spectroscopy (DEIMOS/Keck, MOSFIRE/Keck, EFOSC2/NTT, GMOS/Gemini, Flamingos2/Gemini, VLT/XShooter), optical IFU (MUSE/VLT), mm interferometry (ALMA/NOEMA), JWST NIRISS/NIRCam WFSS and imaging data, JWST NIRSpec IFU
- **Current main developer** and maintainer of interferopy; an open-source, public python library for radio/mm interferometry data analysis.
- **Computing:** Python, C++, Fortran, IDL
- **Astronomy software:** CASA, GILDAS, ESORex, ESOReflex, PypeIt, DS9, Topcat, SourceExtractor, VPFit, GalSim, GALFIT, Grizli, MIRAGE, JWST pipeline, msaexp
- **Summer schools:** European Radio Interferometry School 2019, Chalmers, Gothenburg / Astrostat 2021, Penn State (Virtual) / ISM Galaxies 2021, Marseille (Virtual)

INVITED AND CONTRIBUTED TALKS

Invited Talks and Colloquia:

Miracles of the Early Universe II - Geneva, Switzerland June 2026
Connecting the Sources of Reionisation to the IGM topology
Hamburg Observatory Colloquium - Hamburg January 2025
The ALMA View of Quasars in the First Billion Years
Galaxy Evolution Circle - LAM, Marseille December 2024
Can we solve Reionisation on the scales of individual ionised bubbles?
Cake Talk - DAWN, Copenhagen May 2024
Probing early galaxy formation with an unbiased sample of $6.8 < z < 9$ *[OIII] emitters*
Galaxies Discussion Group - KICC, Cambridge October 2022
A resolved view of $z > 6$ *quasar host galaxies*
Extragalactic Seminar - UCL, London October 2022
A 100-300 pc view of $z > 6.5$ *quasar host galaxies*
Young Astronomers on AGN 2022 - DIPC, San Sebastian October 2022
Quasar Host Galaxies at Cosmic Dawn

Astronomy Seminar - University of Southampton March 2022
Quasar Host Galaxies at Cosmic Dawn
Königstuhl Colloquium - MPIA, Heidelberg April 2021
Finding the sources of reionisation
Galaxies Discussion Group - KICC, Cambridge April 2020
IGM-galaxy cross-correlations during Reionization
MPIA Galaxy Coffee - MPIA, Heidelberg November 2019
Probing reionisation with cross-correlations of galaxies and the Lyman-α forest at $z \sim 6$
London Cosmology Discussion Meeting - RAS, London November 2018
Probing the epoch of reionisation with cross-correlations of high-redshift galaxies and the IGM transmission

Contributed Talks:

MIAPbP Workshop: Continuing the JWST revolution - Munich, DE June 2026
Towards a 3D view of Reionisation
Massive Black Holes across Cosmic Time - Cambridge, UK September 2025
Discovery of two z∼6.5 post-quasar galaxies in large ionised bubbles
EREBUS Collaboration Meeting - Bologna, Italy June 2025
A MUSE Archival View of z∼6 Quasar Environments
Miracles of the Early Universe - Geneva, Switzerland June 2025
Towards a 3D view of Reionisation: Connecting the Sources to the IGM topology
Galaxy Origins in the JWST era - Toledo, Spain May 2025
Discovery of two z∼6.5 post-quasar galaxies in large ionised bubbles
Views on the multi-phase ISM in galaxies - Bologna, Italy September 2024
Pushing ALMA to the limit: 140-pc resolution [CII] and continuum observations of a z=6.6 quasar-galaxy merger
Observing and Simulating Galaxy Evolution - Ascona, Switzerland August 2024
Constraining early galaxy formation with an unbiased sample of $6.8 < z < 9$ *[OIII] emitters*
EAS 2023 - Krakow, Poland July 2023
Contributed talks in Symposium 12, Special Session 22 & Poster Presentation in Symposium 8
First Light Conference,MIT, Boston June 2023
A complete census of [OIII]+Hβ emitters at $6.7 < z < 9$ *with JWST*
Exploring the evolving Universe (RSE@70+) - IoP, London June 2023
A complete census of [OIII]+Hβ emitters at $6.7 < z < 9$ *with JWST*
From Stars to Galaxies II - Gothenburg, Sweden June 2022
Hyper-resolution observations of $z \sim 6.8$ *quasar hosts*
Learning the high-z Universe - Sazerac Online Conferece February 2022
Learning from Quasars with VAEs
Quasars and Galaxies Through Cosmic Time - Chile / Online February 2022
Constraining galaxy overdensities around three $z \sim 6.5$ *quasars with ALMA and MUSE*
RAS Specialist Day - RAS London / Online January 2022
Measuring the contribution of $z \sim 6$ *galaxies to reionisation with galaxy-IGM cross-correlations*
EAS 2021 - Session 2 - Leiden Observatory / Online July 2021
High-redshift quasar hosts viewed by ALMA and NOEMA
EAS 2021 - Session 7 - Leiden Observatory / Online July 2021
Measuring escape fractions at z∼6 with quasar spectra
SAZERAC 2020 - Online July 2020
Measuring the ionising photon escape fraction of $z \sim 6$ *galaxies*
EAS 2020 - Leiden Observatory / Online June 2020
Discovery of a double-peaked Lyman alpha emission in a galaxy at $z = 6.802$
KICC 10th Anniversary Symposium - KICC, Cambridge September 2019

Probing the epoch of reionisation with cross-correlations of high-redshift galaxies and the IGM transmission

EAS 2019 (Session 2) - Lyon, France June 2019

Evidence for quasar evolution: rest-frame UV broad lines shifts at $1.5 < z < 7.5$

EAS 2019 (Session 3) - Lyon, France June 2019

A new route to the contribution to reionisation of subluminous $z \sim 6$ *galaxies*

What matters between galaxies? - Abbazia di Spineto, Firenze, Italy June 2019

Metal-tracing the sources of reionisation

IGM2018: Revealing Cosmology and Reionization History with the Intergalactic Medium - Kavli IPMU, Kashiwa, Japan September 2018

Faint galaxies reionising the IGM at $z \sim 5$*: metal-tracing the sources of reionisation*

OUTREACH

Supervision of 3 high-school interns (1 week) May 2025

Geneva Observatory

Public Talk (French): "La Fin des Âges Sombres" April 24, 2026

Société Astronomique du Valais Romand

Public Talk (French): "La Fin des Âges Sombres" February 7, 2025

Société d'Astronomie du Nord Vaudois

Geneva Observatory Visits 2024 - present

University of Geneva - Visits for general public (adults and children)

Astronomy On Tap London 2020

Co-founder and online host (during Covid-19)

UCL Diploma Club Seminar Organiser 2019 - 2020

Seminar series for graduates of the UCL evening course open to the general public

ORBYTS: Researcher in schools program 2017 - 2019

Outreach programs for highschool pupils of 10 weeks of lectures and research on real data

Mid-Kent Astronomical Society January 31 2020

Outreach talk: Galaxies in the First billion years

Bounce Back RAS200 Project (HMP Brixton) November 4 2019

Outreach to inmates of a London prison - with the RAS and the Bounce Back foundation

UCL Future Frontiers Event Careers Networking Event July 2019

Day of networking/career advice to London pupils from underprivileged background

International Day of Light - UCL May 2018

Day of outreach shows and demonstrations for secondary school pupils

PRESS COVERAGE

- November 2022: "Dans les coulisses des données du télescope spatial James Webb": radio interview and web article with the Swiss public broadcasting organisation RTS on the analysis of early JWST data (article and radio interview in French)
- July 2021: "Cosmic Dawn occurred 250 to 350 million years after Big Bang" press release linked to preprint arxiv.org/abs/2104.08168. Covered by the BBC press/TV/radio, the Guardian, the Daily Mail, the Evening Standard and other specialised media. Live interview with BBC World News TV (24/06/21).
- July 2020: EAS press release "Discovery of a luminous galaxy reionizing the intergalactic medium 13 billion years ago", covered by specialised internet media.